

**Department of Chemical and Biomolecular Engineering
North Carolina State University**

CHE 311-003(Hons)

Transport Processes I

Khan, Fall 2023

Time: 8:30 – 9:45 T Th

Classroom: FWH Rm 2331

Instructor: Saad Khan
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Email: khan@ncsu.edu

Course information will be (1) posted on Moodle and/or (2) emailed to students

Teaching Assistants: Darshana Malusare
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Office Hrs: Mondays/Wednesdays from 4-5:30 pm in EB1 2nd floor atrium

Anicah O'Brien
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Required Text:

Bird, Stewart, Lightfoot, and Klingenberg, Introductory Transport Phenomena, Wiley (2015).

Course purpose:

This course is an introduction to principles of mass, energy, and momentum transfer, which form a theoretical and practical basis for engineering design and thermodynamics. It builds on basic elements presented in CHE 205 and 225, developing a more rigorous and thorough description of mass and energy balance in process systems. Additionally, there may be computational-based problems that require MATLAB skills obtained in CHE 225. It also introduces concepts of microscopic balances, described by differential equations of motion, which are useful in describing a wide variety of transport phenomena. Steady state and non-steady state heat transfer are also presented to describe cooling and heating in process systems. By taking the course, a student would be able to:

- 1) Describe and analyze flows of Newtonian fluids
- 2) Describe and analyze energy and heat transport in fluids, solids and between phases
- 3) Incorporate safety provisions in the analysis and design of transport systems
- 4H) Use vector and tensor concepts and notation in describing fluid conservation equations, and flow types.
- 5H) Use descriptors of simple rheology of Non-Newtonian fluids to solve Bingham and power-law fluid flow.
- 6H) Describe use of approximate methods to solve complex fluid flow such as boundary layer

problem.

7H) Solve flow around objects using stream function analysis.

ChE 311-003 (Hons): POLICIES AND PROCEDURES

Prerequisite:

• An overall GPA of ~3.5 and above; as such I will consider you essentially as “grad” students, with all the responsibilities that come with being a graduate student.

Exams:

- There will be three three-hour exams on the materials covered in this course.
- The exams will have to be scheduled outside of the regular class hours, possibly in the evening. Students who have a conflict should make other arrangements with the instructor.
- All tests will be open-book (some closed book part may be included). *Students may not share notes, books, or any other materials during the open book part of the examination.*
- There are no excuses for missed exams other than those certified by university or medical officers. Students who miss an exam for legitimate reasons will have two choices; they can either have the average of their two exams count as their missed exam or they can take a comprehensive make-up exam at the end of the semester. The latter, if necessary, will be scheduled on a single day for all interested students.

Homework:

- There will be homework assignments approximately every week. Please turn in required homework during lecture hours (unless otherwise noted).
- HW will be submitted as hardcopies, not electronically.
- Late homework will not be accepted.
- All final answers must be boxed. All work leading to the final answer must be shown or credit will not be given. Illegible solutions will not receive credit.
- Regrade requests must be submitted to all 3 instructors via email within 1 week after the homework set was due. Regrade requests received after this time period will not be considered.

Group Solutions:

- You are required to work in groups of three on the homework problems. You can form your own groups; however disproportionate groups will be reassigned by the instructor.
- *If a student's name appears on a solution set, it certifies that the student has participated in deriving the solution and understands all of it.* Try setting up problems individually before getting together.
- Turn in one homework solution set per group with each group member's name listed on the document. Individual homeworks will not be accepted.

In-class Exercises:

- Students will be asked to participate in in-class exercises; these may involve a two-minute answer to solving an entire problem.
- These exercises will be informal with students asking questions while working and sharing answers.
- Instructor may ask individuals or groups to share their answers with the class electronically or orally.
- Participants' names will be noted at the end of class either through a sign up sheet or using Tophat (tophat.com). We will explore the latter as the semester progresses

Dr. Khan's Office Hours:

I will not have formal office hours as I typically have an open door policy. I will also be available by appointment on fairly short notice. I will also be available via email.

Calculation of Final Grade:

- Exams - 75%; • Homework – 15%; • In-class exercises – 10%

Panopto Recordings:

Each class will be recorded via Panopto. Our course folder can be accessed via the following link, <https://ncsu.hosted.panopto.com/Panopto/Pages/Sessions/List.aspx#folderID=6a86f6b1-0470-4330-a953-b05500641ad3>. If you have trouble accessing the recordings, please email help@ncsu.edu for assistance.

Attendance:

Students who miss class due to an excused absence should work with the instructor or problem session TA to make up any missed homework or tests. Documented medical excuses should be presented to the instructor. For a full statement of the university attendance policy, see the following link: www.ncsu.edu/provost/academic_regulations/attend/reg.htm.

Laptop and Cell Phone Use:

Although technology opens up new learning possibilities for students, sometimes students utilize it in ways that are inappropriate. Please refrain from texting, Facebook, instant messaging, e-mailing, surfing the Internet, playing games, etc. during class time since that distracts others and prevents your active participation in the class. Acceptable uses of laptops include taking notes as well as working on assigned in-class activities that require laptop use (most often this will be during problem session). It is easy for your laptop or cell phone to become a distraction to you and to those around you. Please put cell phones on silent when you are in the classroom. *Cell phones must be turned off and put away during exams.*

Academic Integrity

Students should refer to the University policy on academic integrity found in the Code of Student Conduct (found at http://www.ncsu.edu/policies/student_services/student_discipline/POL11.35.1.php). It is the instructor's understanding and expectation that the student's signature on any test or assignment means that the student contributed to the assignment in question (if a group assignment) and that they neither gave nor received unauthorized aid (if an individual assignment). Authorized aid on an individual assignment includes discussing the interpretation of the problem statement, sharing ideas or approaches for solving the problem, and explaining concepts involved in the problem. Any other aid would be unauthorized and a violation of the academic integrity policy.

Documents --lecture notes, lecture videos, homework assignments and solutions, exams and solutions, or handouts --made available to you for this course are intended only for your personal use. You are not allowed to share any content of the class with any person not signed up for the course this semester; a personal, public, or commercial website; or any other news or advertising media.

Supporting Fellow Students in Distress: As members of the NC State Wolfpack community, we each share a personal responsibility to express concern for one another and to ensure that this classroom and the campus as a whole remains a safe environment for learning. Occasionally, you may come across a fellow classmate whose personal behavior concerns or worries you. When this is the case, I would encourage you to report this behavior to the NC State Students of Concern website: <http://studentsofconcern.ncsu.edu/>. Although you can report anonymously, it is preferred that you share your contact information so they can follow-up with you personally.

Census Date (September 2, 2022). Last day to add a course with permission of instructor. Last day for tuition refunds due to dropping a course or changing from credit to audit. Last day for undergraduate students to drop below 12 hours or to drop a course without a W grade.

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Tentative Class Schedule

<u>Week #</u>	<u>Subject</u>	<u>Reading: BSLK</u>
1-2	Introduction , Math Review	Ch. 0, Appendix A
2-3	Fluid Statics Fluid Flow, Shear Stress, Momentum Flux, Newton's Law of Viscosity, Types of Fluid Flow Molecular and Convective Momentum Flux	Ch. 1.1-1.3, 1.7
4	Shell Momentum Balances and Velocity Distributions in Laminar Flow Continuity and Eq of Motion	Ch. 2.1-2.4 Ch. 3.1-3.2
5	Mechanical Energy Balance Substantial Time Derivative Example Problems	Ch. 3.3 Ch. 3.5 Ch. 3.6-3.7
6	Exam 1 tentatively Time Dependent Flow of Newtonian Fluids Stream Functions and Boundary Layers	Ch. 3.8 Notes
7	Flow in tubes, friction factors Macroscopic Mass Balance and Free Jets	Ch. 6.1-6.3 Ch. 7.1-7.2
8	Macroscopic Mech. E. Balance Bernoulli Equation Friction Loss in Piping Systems	Ch. 7.4 Ch. 7.5
9	Macroscopic Balance Problems	Ch. 7.6
10	Thermal Conductivity, Fourier's Law Convective Energy Transport	Ch. 9.2 Ch. 9.1, 9.3, 9.4
11	Newton's Law of Cooling, Shell Energy Balance Internal Heat Generation	Ch. 10.1 Ch.10.2
12	Exam 2 tentatively Conduction Through Composite Walls, Fins, Biot number Forced and Natural Convection	Ch.10.3,5 6
13	Differential Energy Balance	Ch. 11.1-11.4
14	Multiple Variable Temperature Distributions Heat Transfer Coefficients	Ch. 11.5 Ch. 14.1, 3, 4
	Macroscopic Energy Balances	Ch. 15.1-15.3
	Final Exam (Comprehensive)	